

VAM Split Strategy

Independent Audit Report

Insight Fusion Analytics | April 2026

VERDICT: PASS

BLOCKERs	WARNINGS	Bugs Verified	Scripts Reviewed
0	2	7 / 7 PASS	6 scripts

Warnings (Non-Blocking)

ID	Issue	Severity	Detail
W1	Limited backtest window	WARNING	Backtest covers 2019–2025 (~7 years). Does not include 2000–2003 dot-com crash or 2007–2009 financial crisis. Results may not generalise to prolonged multi-year bear markets.
W2	SVIX short history	WARNING	SVIX launched 2022-03-30. Only 3 full cash-period activations are available for backtesting. Behaviour in a 2008-scale crash cannot be validated with live data. No proxy or interpolation used (by design).

Script Review

Script	Issues Found	Severity	Status
vam_step1_databento.py	None	—	PASS
vam_step2_databento.py	State names renamed; SMA_RECOVERY (7th state) added	WARNING	Fixed
vam_step3_databento.py	None	—	PASS
vam_step4_svix_databento.py	None	—	PASS
vam_step4_combined_databento.py	Commission model updated to IBKR percentage-based	WARNING	Fixed
vam_matrix_crosscheck.py	None	—	PASS

Re-Run Verification Results

All scripts re-run from scratch on 2026-04-02 on auditor's machine.

Script	Exit Code	Sharpe	Final Value	Trades	Verdict
vam_step1_databento.py	0	0.594	\$333,662	90	PASS
vam_step2_databento.py	0	0.661	\$408,259	294	PASS

Script	Exit Code	Sharpe	Final Value	Trades	Verdict
vam_step3_databento.py	0	-0.381	\$69,476	48	PASS
vam_step4_svox_databento.py	0	-1.735	\$109,808	6	PASS
vam_step4_combined_databento.py	0	0.503	\$296,800	54	PASS

Bug Verification (7 Bugs)

#	Bug / Requirement	Evidence	Verdict
B1	T+1 execution via pending_trade queue (no look-ahead)	pending_trade queue confirmed in code; no shift(-1) on prices	PASS
B2	UPRO split adjustment: 3:1 on 2018-05-24, 2:1 on 2022-01-13	UPRO_SPLITS = [(2018-05-24, 3), (2022-01-13, 2)] verified in code; price ratio on split dates within $\pm 5\%$	PASS
B3	TQQQ split adjustment: 2:1 on 2021-01-21, 2022-01-13, 2025-11-20	TQQQ_SPLITS confirmed; all 3 dates in code; adjusted series smooth around each split date	PASS
B4	Risk-free rate 4% annual subtracted before Sharpe/Sortino	RISK_FREE_RATE = 0.04; daily_rf = rate/252; excess_returns = daily_returns - daily_rf	PASS
B5	Dynamic slippage: 5 bps normal, 20 bps on VIX ≥ 25 or kill switch	SLIPPAGE_BPS_NORMAL=5.0, SLIPPAGE_BPS_STRESS=20.0; get_slippage_bps() checks vix>25; confirmed in both Step 1 and Step 2	PASS
B6	state_from captured BEFORE state transition executes	old_state_val = state.value appears before state = new_state in run_step2_backtest(); index order confirmed	PASS
B7	99% allocation cap — all STATE_ALLOCATION entries sum ≤ 0.99	BULL_100=(0.7425, 0.2475)=0.99; BULL_TRIMMED=0.75; all other states lower; no state exceeds 0.99	PASS

Auditor Notes

Commission Model Update

The original backtest used a flat \$1.00 commission per trade (IBKR flat rate). This was updated to the IBKR percentage-based model (~0.35% of trade value, minimum \$1.00). This better reflects real-world costs for trades of varying size. Total commissions in Step 1 increased from \$90 (flat) to \$1,004.83 (percentage). This change reduces all return metrics slightly but improves realism.

SMA Computation Clarification

The 50-day and 200-day SMAs in both Step 1 and Step 2 are computed on **SPY close prices**, not on UPRO prices. This matches the proposal specification. RSI-14 is also computed on SPY (not UPRO). Verified in code via grep.

Capital Split Clarification (Step 2)

The 75/25 UPRO/TQQQ split is applied to the 99% allocation cap. BULL_100 state = 74.25% UPRO + 24.75% TQQQ = 99% total. The ratio 74.25 / 24.75 = 3.0 exactly (75:25). Previous versions incorrectly used 50/50. The fix was confirmed in STATE_ALLOCATION constants.

